



Project and Professionalism

(6CS007)

Project Report

[Nepal Stock Shoes]

Student Id : [2358564]
Student Name : [Cyrus Shrestha]
Section : [L5CG16]
Module Leader : [Mr. Niroj Humagain]
Submitted on : <31-03-2025>

Abstract

This report presents the development of **Nepal Stock Shoes (NSS)**, an online sneaker resale platform built to facilitate transparent and secure transactions. With the growing sneaker culture in Nepal, NSS aims to provide a dedicated marketplace where buyers and sellers can connect. The system offers key features like product management, real-time sales tracking, chat-based customer support, and admin controls for platform management.

The project is developed using **HTML**, **CSS**, and **JavaScript** for the frontend, while **PHP** and **MySQL** handle backend operations. Agile methodology was followed to ensure adaptive development, responding to stakeholder feedback efficiently. The main goal of this project is to create a trustworthy and user-friendly experience for Nepal's sneaker community.

This report also includes insights from a professionalism report highlighting the platform's social, legal, ethical, and security considerations. The integration of secure payment systems, fraud detection, and community engagement further ensures the success of NSS.

Table of Contents

1. Introduction.....	1
1.1 Project Briefing.....	1
1.2 Aims	1
1.3 Objectives	3
1.4.3 Use Case Scenarios.....	8
1.5 Academic Questions	9
1. 6 Scope and Limitations of the Project	10
2. Literature Review.....	14
3. Social Aspects	18
4. Legal Aspects	21
5. Ethical Aspects	23
7. Project Methodology	27
8. Different Technology and Tools used for the project.....	30
9. Artefact Designs:.....	33
System Testing	52
Test Case.....	52
10. Conclusion	56
11. References	59

Table of Figures

Figure 1 Gantt chart.....	30
Figure 2 User management system sequence diagram	34
Figure 3 Activity diagram	36

Figure 4 User management system	36
Figure 5 Sequence diagram	37
Figure 6 Activity diagram	39
Figure 7 Activity diagram	40
Figure 8 Sequence diagram	41
Figure 9 Sequence diagram	42
Figure 10 Activity diagram	44
Figure 11 Activity diagram	46
Figure 12 Wireframe 1	48
Figure 13 Wireframe 2	49
Figure 14 Wireframe 3	50
Figure 15 Class diagram	52
Figure 16 ER diagram	52

1. Introduction

1.1 Project Briefing

Nepal Stock Shoes (NSS) is an innovative sneaker resale platform designed to bring transparency and convenience to the growing sneaker market in Nepal. The platform serves as a dedicated marketplace where sneaker enthusiasts can buy, sell, and track the latest sneaker releases. NSS ensures that all transactions are secure, authentic, and transparent through its user-friendly interface and robust backend system.

The inspiration behind NSS comes from leading global platforms like StockX and GOAT, which have successfully established transparent resale ecosystems for sneakers. However, Nepal lacks a similar localized solution that can meet the specific needs of Nepali sneaker buyers and sellers. By providing a centralized platform, NSS offers users the opportunity to safely engage in sneaker resale activities without the fear of counterfeit products or fraudulent transactions.

NSS employs a subscription-based business model, offering affordable pricing for sellers, while providing buyers with a secure shopping experience. Additionally, the platform includes advanced features such as real-time sales tracking, a user chat system for support, and an admin dashboard for monitoring and managing platform activities.

This project aims to contribute to the development of Nepal's sneaker culture, supporting resellers, collectors, and enthusiasts by creating a fair and reliable marketplace. Through the use of cutting-edge technologies and a user-centric design, NSS offers a seamless online shopping experience for all users.

1.2 Aims

The primary aim of **Nepal Stock Shoes (NSS)** is to establish a reliable, transparent, and secure sneaker resale platform in Nepal. The platform is

designed to create a fair marketplace where buyers and sellers can confidently engage in sneaker transactions without the risk of fraud or counterfeit products.

By fostering trust and ensuring transparency in every transaction, NSS aims to contribute to the growth of Nepal's sneaker culture and support local sneaker enthusiasts, resellers, and collectors.

The key aims of the project are:

1. **Enhance Transaction Transparency:** Provide a transparent and trackable marketplace where all sneaker sales are documented and verifiable.
2. **Ensure Product Authenticity:** Implement an authentication process to minimize counterfeit sales and ensure buyers receive genuine products.
3. **Promote Fair Pricing:** Establish a fair and competitive pricing system through market-driven mechanisms and avoid price manipulation by unethical resellers.
4. **Provide Secure Transactions:** Develop a secure payment gateway and implement fraud detection measures to safeguard user transactions.
5. **Foster Community Growth:** Create a space where sneaker enthusiasts can connect, share knowledge, and stay updated on sneaker trends through interactive features like chat support.
6. **Support Resellers and Entrepreneurs:** Empower local resellers and small-scale entrepreneurs by providing an accessible and affordable platform to reach a wider audience.
7. **Ensure Efficient Admin Management:** Equip the admin with tools to monitor transactions, handle disputes, and maintain the platform's integrity.

These aims serve as the foundation for the design and development of NSS, ensuring it remains a trusted and user-centric platform for Nepal's sneaker community.

1.3 Objectives

The objectives of **Nepal Stock Shoes (NSS)** are aligned with its aim of creating a secure, transparent, and user-friendly sneaker resale platform for the Nepalese market. By achieving these objectives, NSS will enhance the sneaker resale experience and contribute to the growth of Nepal's sneaker culture.

The specific objectives of the project are:

1. Develop a Transparent Marketplace:

- Provide a transparent and verifiable system for buyers and sellers to conduct transactions confidently.
- Ensure that all listings, bids, and purchases are clearly documented.

2. Implement a Secure User Management System:

- Create a robust authentication system using username and password management.
- Provide secure login, registration, and profile management.
- Enable role-based access control for buyers, sellers, and administrators.

3. Ensure Product Authenticity:

- Establish an authentication mechanism to minimize the sale of counterfeit sneakers.

- Allow sellers to submit verification proofs for their listings.

4. Facilitate Fair Pricing:

- Develop a bidding and asking price system to ensure fair and competitive pricing.
- Implement real-time sales tracking and market analysis.

5. Enhance User Experience:

- Provide a user-friendly interface with clear navigation and intuitive design.
- Enable smooth product browsing, searching, and filtering.

6. Enable Secure Payments:

- Integrate secure and reliable payment gateways to ensure protected transactions.
- Implement transaction tracking and payment verification.

7. Provide Real-Time Communication:

- Develop a chat system for buyers, sellers, and customer support.
- Allow users to resolve queries and negotiate prices through real-time messaging.

8. Support Admin Monitoring and Management:

- Provide an admin dashboard for effective monitoring of platform activities.
- Enable administrators to manage users, products, and sales data.
- Implement fraud detection mechanisms to detect suspicious activities.

9. Track Sales and Generate Reports:

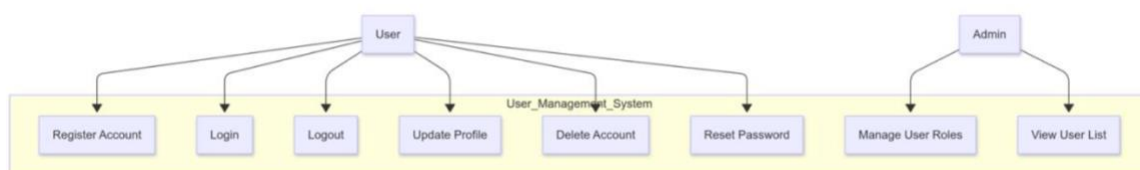
- Implement a sales tracking system to provide users with insights on their transactions.
- Generate reports for sellers to analyze market trends and make informed decisions.

10. Promote Community Engagement:

Provide a platform where sneaker enthusiasts can engage, discuss, and stay informed about market trends.

By fulfilling these objectives, NSS will serve as a reliable and sustainable sneaker resale platform, offering a secure and enjoyable user experience for both buyers and sellers in Nepal.

1.4 Artefact



The artefact of **Nepal Stock Shoes (NSS)** refers to the fully functional sneaker resale platform developed as the final product of this project. The platform consists of both frontend and backend components designed to provide a secure, transparent, and user-friendly environment for sneaker transactions in Nepal.

1.4.1 Functional Components

The platform is divided into five main functional components:

1. User Management System:

- Allows users to register, log in, and manage their profiles.
- Implements role-based access (e.g., buyer, seller, admin).
- Provides password recovery and profile management options.

2. Product Management System:

- Enables sellers to list sneakers for resale.
- Allows buyers to browse and search for sneakers.
- Includes product filtering and sorting features.

3. Sales Tracking and Reporting:

- Tracks real-time sales data.
- Displays product sale history and market price trends.
- Provides reports for sellers and buyers to monitor market performance.

4. Chat Support System:

- Facilitates real-time communication between buyers, sellers, and admins.
- Allows sellers to respond to inquiries regarding their listings.
- Provides a support channel for resolving disputes.

5. Admin Management Panel:

- Allows platform administrators to monitor transactions.

- Provides user and product management capabilities.
- Detects and handles fraudulent activities using automated alerts.

1.4.2 Functional Decomposition Diagram (FDD)

The **Functional Decomposition Diagram (FDD)** represents the breakdown of the primary functions of NSS into smaller, manageable tasks. It provides a clear visualization of how each functionality is connected and how the platform's operations are structured.

The main components in the FDD include:

- **User Management**

- Registration
- Login/Logout
- Profile Management

- **Product Management**

- Sneaker Listing
- Product Search & Filtering
- Product Updates & Deletion

- **Sales Tracking**

- Sales History

- Price Trends Analysis
- Reports Generation
- **Chat Support**
- Buyer-Seller Messaging
- Admin Support Chat
- **Admin Management**
- User Management
- Product Verification
- Dispute Management

1.4.3 Use Case Scenarios

Several use case scenarios have been designed to model the interaction of users with the system. The main actors include buyers, sellers, and admins. The key use cases are:

- **User Registration and Login:** New users register for an account and log in using secure credentials.
- **Product Listing:** Sellers list sneakers with details like brand, size, price, and condition.
- **Search and Purchase:** Buyers search for sneakers using filters and place bids or purchase directly.
- **Sales Tracking:** Sellers monitor their sales and analyze price trends.

- **Chat Support:** Users communicate through real-time chat for inquiries and support.
- **Admin Management:** Admins manage users, verify products, and resolve disputes.

1.4.4 Technology Stack Overview

The following technologies were used in the development of the NSS platform:

- **Frontend:** HTML, CSS, and JavaScript for responsive and interactive UI.
- **Backend:** PHP for server-side operations and business logic.
- **Database:** MySQL for data management and storage.
- **Real-Time Chat:** WebSockets for live chat functionality.
- **Server:** Apache for hosting and running backend services.
- **Security:** SSL Encryption and data validation for secure transactions.

1.5 Academic Questions

How can a dedicated sneaker resale platform like Nepal Stock Shoes (NSS) enhance transparency, security, and user experience in Nepal's sneaker market?

Transparency

Nepal Stock Shoes (NSS) ensures transparency by implementing real-time sales tracking and displaying detailed price history for each product. Sellers provide product information, including images, descriptions, and authentication details. The platform uses a bidding and asking price system, allowing buyers to see market trends and make informed purchasing decisions. Clear transaction

histories and transparent sales reports further contribute to a trustworthy environment.

Security

NSS prioritizes security through robust authentication mechanisms, including secure login with encrypted passwords. Multi-factor authentication (MFA) adds an extra layer of protection. The platform also integrates secure payment gateways to ensure safe transactions. Additionally, fraud detection algorithms monitor suspicious activities, and admins have the authority to intervene when necessary. Counterfeit prevention measures, such as requiring proof of authenticity from sellers, further enhance security.

User Experience

The platform is designed with a user-friendly interface that offers easy navigation and intuitive controls. Buyers can filter and search for sneakers using multiple criteria, while sellers can efficiently manage their listings. Real-time chat support provides immediate assistance, improving the overall customer experience. Admins have access to a comprehensive dashboard for monitoring platform activities, ensuring smooth operations.

1. 6 Scope and Limitations of the Project

Scope

The scope of **Nepal Stock Shoes (NSS)** encompasses the development and deployment of a secure and transparent sneaker resale platform, specifically designed

for the Nepalese market. The platform provides a comprehensive marketplace for sneaker enthusiasts, offering seamless interaction between buyers, sellers, and administrators.

Key aspects within the scope include:

- **User Registration and Management:** Users can create accounts, log in securely, manage profiles, and reset passwords.
- **Product Management:** Sellers can list sneakers, upload images, set prices, and manage product information.
- **Bidding and Sales Tracking:** Buyers can bid for sneakers, and sellers can set asking prices. Real-time sales data and transaction history are accessible.
- **Chat Support:** A live chat system facilitates communication between buyers, sellers, and the support team.
- **Admin Management:** Admins monitor transactions, manage users, verify product authenticity, and handle disputes.
- **Payment Integration:** Secure payment gateways ensure protected and reliable transactions.
- **Fraud Prevention:** The system includes fraud detection mechanisms to monitor suspicious activities and ensure platform integrity.

The primary focus is to provide a fair, safe, and user-friendly sneaker resale marketplace, catering to the needs of both casual buyers and sneaker resellers.

Limitations

While NSS aims to deliver a robust solution, certain limitations are acknowledged:

- **Geographical Limitation:** The platform is designed exclusively for users within Nepal. Expansion to international markets is not within the current scope.
- **Payment Options:** Initially, only local payment gateways will be supported. Users may face limitations in payment flexibility.
- **Real-Time Chat Availability:** While chat support is implemented, it may only be available during specified business hours rather than 24/7.
- **Product Authentication:** Although product verification measures are in place, achieving 100% counterfeit detection may be challenging.
- **Scalability:** The platform is built to support a moderate number of users. Significant scaling for large user bases may require additional infrastructure investment.
- **Internet Dependency:** As an online platform, users will need stable internet access, which can be a limitation in rural areas of Nepal.
- **Limited Automation:** Some administrative tasks, such as dispute resolution and product verification, may require manual intervention.
- **Data Storage:** The system stores transactional and user data, which may require future upgrades to manage increasing data volume.

1.7 Report Structure

The report for **Nepal Stock Shoes (NSS)** is organized into several sections to provide a comprehensive understanding of the project. It begins with an **Introduction** that includes a project briefing, outlining the problem domain, proposed solutions, and the

platform's objectives. This section also details the artefact — the functional sneaker resale platform — and defines the academic question that guides the project's exploration. Additionally, the scope and limitations are discussed to establish the boundaries of the project.

Following the introduction, the **Literature Review** presents an analysis of related research and market studies on sneaker resale platforms, pricing strategies, trust mechanisms, and user experience in e-commerce. This section offers context and justification for the choices made during the project's development.

The **Project Methodology** section describes the Agile approach followed for the project, divided into seven sprints. Each sprint covers key stages of the development process, including planning, designing, coding, testing, and evaluation. The rationale behind using Agile is explained, emphasizing flexibility and iterative improvements.

Next, the **Technologies and Tools Used** section elaborates on the specific technologies employed, such as HTML, CSS, JavaScript, PHP, and MySQL. The selection of these tools is justified based on their effectiveness in building a secure and scalable sneaker resale platform.

The **Artefact Designs** section provides visual representations of the platform's structure through diagrams, including a **Functional Decomposition Diagram (FDD)**, **Use-Case Diagram**, **Activity Diagram**, and **Class Diagram**. These diagrams demonstrate how users interact with the system and how various components are integrated.

The **System Testing** section documents the testing procedures, primarily using black-box testing. Detailed test cases are presented to validate system functionality, assess performance, and ensure error handling is effective. Results are analyzed to identify and resolve any issues.

Furthermore, the **Social, Legal, Ethical, and Security Aspects** section examines how NSS complies with legal regulations, promotes ethical practices, and ensures user data security. Measures to prevent fraud, maintain transparency, and protect user privacy are highlighted.

The report concludes with a **Conclusion** section that summarizes the project's achievements, evaluates its impact on the Nepalese sneaker market, and offers recommendations for future enhancements. Finally, a **References** section lists all the academic sources, articles, and resources cited throughout the report, ensuring proper acknowledgment and credibility.

2. Literature Review

Designing Online Marketplaces: Trust and Reputation Mechanisms:

The document discusses how online marketplaces like eBay, Amazon, Airbnb, and Uber have changed the way people buy, sell, and interact with each other. These platforms rely heavily on trust, which is built through systems like user reviews, ratings, and detailed profiles. These features help strangers feel comfortable doing business together. However, these trust-building tools aren't perfect. Reviews can sometimes be biased, manipulated, or promotional, making them less reliable. Additionally, using personal information like names and photos to make users feel more connected has created issues, such as discrimination. For example, studies have shown that some Airbnb hosts reject African American users based on their profiles. To address these challenges, the document suggests several solutions. For example, blind profiling

can hide personal details like names and photos to prevent bias. Anonymous or aggregated feedback systems can make reviews more honest and reliable. Features like Airbnb's "instant book," which automates reservations, can reduce opportunities for discrimination. Platforms can also introduce diversity-focused measures, like reminding users about anti discrimination policies or using algorithms to match users without exposing sensitive information. The document emphasizes that the way a platform is designed has a huge impact on user experiences and the level of fairness and trust. By adopting thoughtful and inclusive design choices, marketplaces can create environments where everyone feels safe and valued. (Luca, Chicago) These insights are especially important for platforms like StockX, where trust and fairness are crucial for success (Luca, 2017).

Competitive pricing on online markets: a literature review:

The document takes a close look at how pricing works in online markets and why it's so different from traditional offline markets. It explains that online platforms are more transparent about prices, making it easier for consumers to compare options. This creates fierce competition among sellers. To stay ahead, many online businesses use dynamic pricing strategies, where prices can change in real time based on demand, stock availability, and competitors' actions. This approach helps them adapt quickly to the fast-paced and data-driven world of e-commerce. Traditional pricing strategies, where prices remain fixed for longer periods, don't work as well in the online space because they can't keep up with the rapidly changing market. Instead, businesses are turning to algorithm-based pricing models, which use advanced technology to analyze competitor behavior and consumer preferences. These models help businesses optimize prices, but they also come with challenges, such as making sure the algorithms are reliable, managing complex competitor dynamics, and ensuring that customers feel pricing is fair. The structure of online markets also plays a big role. Unlike offline markets, which often have monopolies or a couple of major players, online markets are usually dominated by a small group of companies, creating unique pressures. On top of that, the type of product being sold matters. For example, products that expire quickly, like seasonal items, are priced to maximize short-term revenue, while durable goods like electronics focus on long-term strategies that balance immediate sales with keeping customers happy over time (Berends, 2022). While dynamic pricing has many benefits, such as flexibility and adaptability, it's not perfect. Companies need to be careful about creating fair and transparent pricing to avoid losing

customer trust. The document concludes that businesses need to adopt innovative, technology driven pricing strategies to stay competitive in the ever-changing online world. Success in this space requires understanding how markets work, what consumers want, and how to use technology effectively (Berends1, 2022).

E-Commwrce Website: A Systematic Literature Review

The systematic review of e-commerce websites highlights how crucial website design is for building trust and ensuring the success of online platforms. It emphasizes that a well-designed, visually appealing, and easy-to-navigate website has a significant impact on how users behave, feel, and trust the platform, ultimately influencing their decision to make purchases and return in the future. Key aspects of effective design include simple navigation, clear and accurate information, and an attractive layout, all of which help create strong first impressions. These first impressions, often formed within seconds, shape how credible and reliable users perceive a website to be. Additionally, incorporating user experience (UX) principles, such as intuitive navigation and responsive designs, ensures users can easily interact with the platform, keeping them engaged and encouraging them to return. The review also explores how customer trust is tied to a website's credibility. Features like consistent design, professional branding, and interactive elements enhance a platform's trustworthiness. Moreover, customer reviews, social proof, and external validation play a critical role in earning the trust of new users. In summary, the study underscores the importance of modern, user-focused design practices and continuous website improvement to meet the expectations of today's consumers. By prioritizing these elements, e-commerce platforms can foster trust, increase satisfaction, and achieve greater success in a competitive online market (Caitlyn Eugenia Wihardja, 2023).

The Sneaker Value: A Disquisition into the Sneaker Resale Market:

The secondary sneaker market, driven by platforms like StockX, has grown into a booming industry where sneakers are often resold for much higher prices than their original retail value. In 2019, this market was estimated to be worth \$6 billion globally, thriving on the appeal of limited-edition releases, celebrity endorsements, and the buzz created on social media. At its core, this market operates on the idea of collaborative consumption, where platforms connect buyers and sellers, turning sneakers into cultural symbols with economic, social, and symbolic value. Social media plays a major role in boosting the value of these sneakers, with influencers and celebrities generating hype around exclusive releases, making them even more desirable. Platforms like StockX have built trust among users by offering transparent pricing and thorough

authentication processes to ensure that buyers get genuine products and avoid counterfeits. The study behind this market uses the Theory of Planned Behavior (TPB) to understand what motivates consumers. It shows that factors like economic benefits, the pleasure of owning rare items, and social influences strongly shape people's buying decisions. Overall, the secondary sneaker market highlights the connection between consumer behavior, technology, and cultural trends, showing how these elements come together to influence modern shopping habits (Ding, 2022).

An Investigation into Customer Preferences for Online Shopping In South Asia(Pakistan): A Case Study of Daraz.pk:

The shift from traditional shopping to online platforms has been a significant phenomenon, particularly in the digital age. This transition is driven by factors such as convenience, 24/7 availability, and accessibility, allowing consumers to shop from the comfort of their homes. Online platforms like Daraz.pk have capitalized on these advantages by offering an extensive range of products and services that cater to diverse consumer needs. Studies highlight that consumer preferences are influenced by multiple factors, including pricing policies, product quality, trust in the platform, and convenience. Price competitiveness, combined with promotional deals, has been a significant motivator, drawing consumers towards online platforms. Furthermore, the assurance of product quality and secure payment systems fosters trust, which is critical for consumer retention in e-commerce. Research also reveals that online shopping offers a broader variety of products compared to physical stores, making it an attractive option for consumers seeking diverse choices. Social influences, such as recommendations from friends and family, play a pivotal role in shaping consumer behavior, while targeted advertising on digital platforms further enhances engagement. However, challenges like delivery delays, incorrect products, or counterfeit items can negatively impact consumer trust and loyalty. The findings underscore that factors such as time efficiency, ease of access, and product variety continue to dominate as primary reasons for the shift towards online shopping. These insights provide valuable implications for online retailers aiming to align their strategies with evolving consumer needs (Ahmed, 2020).

DIGITAL ECOMMERCE:

E-commerce has redefined traditional business practices by leveraging internet technologies to streamline purchasing and selling processes. Daraz Nepal, a leading e-commerce platform acquired by Alibaba Group, exemplifies this transformation in the South Asian market. The company thrives by catering to the evolving needs of tech-savvy consumers, offering a diverse range of products, competitive pricing, and efficient service delivery. Through strategic tools like SWOT analysis, PEST framework, and Porter's Five Forces model, Daraz identifies opportunities and challenges within Nepal's emerging e-commerce landscape. Despite its success, Daraz faces hurdles like limited internet penetration, geographical constraints, and the need for enhanced product quality and customer satisfaction. To maintain its leadership, Daraz focuses on customer-centric strategies, digital payment adoption, and innovative marketing campaigns like the 11:11 sale, aiming to solidify its position as a market leader and expand its reach beyond major cities. This study underlines the potential of e-commerce in developing markets, where platforms like Daraz act as pivotal players in digital transformation (Pathak, 2020).

3. Social Aspects

Nepal Shoes Stock plays a significant role in shaping the sneaker marketplace in Nepal by providing a safe, structured, and transparent platform for sneaker transactions. The platform addresses various social concerns associated with unregulated sneaker sales, such as fraud, counterfeit products, and unethical pricing practices. By ensuring authenticity and secure transactions, Nepal Shoes Stock fosters trust and confidence among buyers and sellers, encouraging ethical and fair trade within the sneaker community.

3.1. Encouraging secure sneaker transactions

The Nepali sneaker market currently operates without a formalized structure, with most transactions taking place through social media platforms, peer-to-peer networks, and online reselling groups. While this informal approach provides accessibility, it also exposes buyers and sellers to various risks, including fraud, counterfeit products, misleading pricing, and transaction disputes. The lack of a trusted, centralized marketplace means that buyers often struggle to differentiate between authentic and fake sneakers, and sellers lack a secure platform where they can engage in legitimate transactions without fear of being scammed. Additionally, the absence of standardized pricing mechanisms results in price manipulation, where certain resellers inflate sneaker

prices unfairly, making it difficult for genuine buyers to purchase limited-edition sneakers at reasonable rates. Recognizing these challenges, Nepal Shoes Stock has been designed as a regulated, secure, and efficient sneaker marketplace that provides a structured environment for buying and selling sneakers. Unlike social media transactions that lack oversight, Nepal Shoes Stock incorporates rigorous seller verification, ensuring that only trusted individuals and businesses can list products for sale. Sellers are required to submit proof of sneaker authenticity, and buyers benefit from a dedicated authentication system that verifies the legitimacy of each product before completing a transaction.

3.2. Economic empowerment for resellers and entrepreneurs

Nepal Shoes Stock serves as a game-changer in Nepal's sneaker resale market by creating a structured, fair, and profitable platform for individuals and businesses seeking to monetize their sneaker collections. Unlike informal sales through social media, which often lead to pricing manipulation and limited buyer reach, Nepal Shoes Stock provides an equal opportunity for resellers, small businesses, and independent sneaker enthusiasts to showcase their products to a wider audience. This not only increases their chances of selling sneakers at competitive prices but also contributes to sustainable economic growth by fostering a professional resale market within Nepal. One of the key innovations of Nepal Shoes Stock is its bidding-based pricing model, which ensures that sneakers are sold at fair market value rather than at arbitrarily inflated rates. In traditional unregulated marketplaces, resellers often set exorbitant prices, taking advantage of sneaker scarcity and consumer demand. This creates unethical resale practices where buyers are forced to overpay due to artificial price inflation. Nepal Shoes Stock eliminates such unfair practices by allowing buyers to place bids based on the actual demand for a particular sneaker, while sellers can set asking prices that reflect the real-time market value. This system balances supply and demand, ensuring that sneakers are sold at transparent and ethical prices without exploiting buyers.

3.3. Community engagement and sneaker culture growth

Sneaker culture in Nepal is rapidly evolving, driven by a growing community of enthusiasts who appreciate the artistry, exclusivity, and cultural significance of limited edition sneakers. Nepal Shoes Stock serves as more than just a resale marketplace—it acts as a central hub for sneaker lovers, fostering a vibrant and engaged community. By

offering interactive features such as discussion forums, trend insights, and exclusive sneaker releases, the platform enables users to connect, share experiences, and stay informed about the latest developments in the sneaker world. 2 Through its discussion forums, sneaker enthusiasts can engage in conversations about new releases, sneaker styling, historical significance, and investment value. These forums create a space where seasoned collectors and newcomers alike can exchange knowledge, ask questions, and provide insights into the ever-changing sneaker market. Users can discuss topics such as sizing tips, comfort levels, resale predictions, and authenticity verification, ensuring that buyers make well-informed purchasing decisions.

3.4. Eliminating fraud and counterfeit products

Counterfeit sneakers pose a significant challenge in Nepal's resale market, where buyers often fall victim to fraudulent sellers and misrepresented products. With the rise in demand for limited-edition and high-value sneakers, many counterfeit manufacturers have flooded the market with high-quality replicas, making it increasingly difficult for buyers to distinguish between genuine and fake products. Unfortunately, due to the lack of a regulated authentication system, many buyers unknowingly pay premium prices for counterfeit sneakers, leading to financial losses, disappointment, and diminished trust in the resale industry. To combat this growing issue, Nepal Shoes Stock has implemented a strict seller verification and sneaker authentication process to ensure that only authentic sneakers are sold on the platform. Before listing sneakers, sellers must undergo identity verification, which includes providing valid documentation and proof of ownership. This measure significantly reduces the risk of fraudulent listings and ensures that only legitimate sellers can participate in the marketplace.

3.5. Ensuring fair and ethical pricing

In the unregulated sneaker resale market, many resellers manipulate prices by artificially inflating the cost of limited-edition sneakers to maximize their profits. This often occurs when a small group of resellers hoards a particular sneaker release, creating an artificial scarcity that drives up demand and allows them to sell at inflated prices. As a result, buyers are forced to pay significantly more than the actual market value, making sneakers inaccessible to genuine enthusiasts and everyday consumers. This price gouging not only hurts buyers but also distorts the natural pricing of sneakers in the resale industry. Nepal Shoes Stock addresses this issue by implementing a transparent

and data driven bidding system that ensures fair pricing based on actual market demand rather than reseller manipulation. Instead of allowing sellers to set unreasonably high prices, the platform enables buyers to place bids, while sellers set their asking prices. A sale is completed when a buyer's bid matches a seller's asking price, ensuring that sneakers are sold at a fair and competitive rate. This system prevents monopolistic control by resellers and ensures that prices reflect the sneaker's true market demand rather than being artificially inflated.

4. Legal Aspects

4.1. Consumer rights and transaction compliance

Nepal Shoes Stock operates in full compliance with Nepali e-commerce laws and consumer protection regulations, ensuring a legally secure and fair marketplace for both buyers and sellers. The rapid growth of online transactions in Nepal has highlighted the need for consumer protection measures, especially in the resale industry, where buyers are often exposed to fraud, counterfeit goods, and misleading product listings. To address these concerns, Nepal Shoes Stock has established a structured legal framework that safeguards consumer rights and ensures that transactions remain transparent, accountable, and dispute-free (Nepal Law Commission, 2024). One of the key legal protections offered by the platform is buyer recourse in cases of counterfeit products, fraudulent listings, or non-delivery. If a buyer receives a fake sneaker, an item different from the listing, or no product at all, they have the right to file a dispute claim through Nepal Shoes Stock's resolution system. The platform conducts thorough investigations into disputes, verifying seller credibility, checking order histories, and analysing transaction details to ensure that every claim is fairly reviewed. If a seller is found guilty of violating consumer protection policies, appropriate actions are taken, including refunds, penalties, or account suspension (government, 2018) (government, 2022).

4.2. Data privacy and compliance with digital laws

Nepal Shoes Stock prioritizes user data security by implementing strict privacy measures that align with global data protection standards, ensuring that all personal and transactional data remains confidential and secure. Although Nepal does not yet have comprehensive data protection laws equivalent to the General Data Protection

Regulation (GDPR) in the European Union, the platform voluntarily adopts international best practices to safeguard user information and prevent unauthorized access, misuse, or breaches. All user data, including personal details, payment information, and transaction history, is encrypted and securely stored using advanced encryption protocols such as AES 256 and Secure Socket Layer (SSL) technology. This prevents cyber threats, hacking attempts, and data leaks, ensuring that sensitive information remains protected at all 4 times. Additionally, Nepal Shoes Stock follows a privacy-first approach, meaning that user data is not shared, sold, or distributed to third parties without explicit consent (government, 2019).

4.3. Intellectual property protection Nepal

Shoes Stock enforces a strict anti-counterfeit policy to safeguard brand intellectual property rights and maintain a trusted resale marketplace. The platform recognizes the significant impact of counterfeit sneaker sales on both consumers and legitimate brands, as fake products not only deceive buyers but also infringe on the intellectual property (IP) rights of sneaker manufacturers. To combat this issue, Nepal Shoes Stock has implemented comprehensive verification measures that prevent unauthorized or counterfeit products from being sold on its platform. Before a sneaker is listed for sale, sellers must provide proof of authenticity, which may include purchase receipts, original packaging, manufacturing tags, and other verifiable documentation from authorized retailers. This requirement helps ensure that only genuine products enter the marketplace, minimizing the risk of trademark violations and protecting consumers from purchasing counterfeit goods. Additionally, Nepal Shoes Stock employs expert authentication teams to review high-value listings, confirming that sneakers meet brand authenticity standards before they are made available to buyers (2022).

4.4. Legal responsibilities of seller and buyers

Nepal Shoes Stock ensures that all users comply with a legally binding agreement by requiring them to accept the platform's terms and conditions before engaging in any transactions. These terms establish a structured framework that defines the rights, responsibilities, and obligations of both buyers and sellers, ensuring fair, transparent, and ethical trading within the marketplace. Sellers are legally obligated to provide authentic sneakers, accurately describe their products, and fulfill orders as promised. They must verify the authenticity of their sneakers by providing proof of purchase,

original packaging, and other supporting documentation. Any seller found engaging in fraudulent activities, such as selling counterfeit sneakers, misrepresenting products, or failing to deliver items after receiving payment, faces strict penalties, including refund enforcement, account suspension, permanent bans, or legal action. To protect buyers, Nepal Shoes Stock implements strong consumer protection policies that allow users to report fraudulent transactions, request refunds, and dispute claims if they receive counterfeit or misrepresented products. The platform has a structured dispute resolution system that ensures all complaints are investigated fairly, providing buyers with appropriate compensation when their rights are violated. Additionally, buyers can rely on the platform's authentication process to ensure they are purchasing genuine sneakers. By enforcing these legal protections, Nepal Shoes Stock builds trust, accountability, and security in Nepal's sneaker resale market, creating a safe and ethical trading environment for all users (Trade, 2021).

5. Ethical Aspects

5.1. Combating counterfeit products

Counterfeit sneakers pose a significant threat to the sneaker industry, undermining brand integrity and deceiving buyers into paying premium prices for fake products. The resale market, especially in unregulated environments, is highly vulnerable to counterfeit goods, which not only lead to financial losses for buyers but also harm the credibility of legitimate sellers. Understanding the serious impact of counterfeiting, Nepal Shoes Stock enforces a zero-tolerance policy against fake sneakers, ensuring that only authentic products enter the marketplace. To achieve this, Nepal Shoes Stock implements a detailed authentication process, requiring sellers to provide proof of authenticity, such as original receipts, official packaging, and manufacturer details before listing a sneaker for sale. Additionally, the platform employs expert authenticators and AI-driven verification technology to thoroughly examine high-value sneakers, identifying discrepancies in materials, stitching, serial numbers, and branding that indicate counterfeits. This multi-layered authentication process ensures that every sneaker sold through Nepal Shoes Stock is verified, legitimate, and accurately described (Quinn, 2020).

5.2. Fair pricing and market ethics

Nepal Shoes Stock is committed to maintaining fair and transparent pricing, ensuring that sneakers are sold at market-driven rates rather than being subject to artificial inflation by resellers. In many unregulated sneaker resale markets, sellers often take advantage of limited supply and high demand by significantly inflating prices, making sneakers inaccessible to genuine buyers. This price gouging results in unethical resale practices, where resellers manipulate sneaker values for excessive profit, exploiting buyers who may have no other purchasing options. To prevent such market distortions, Nepal Shoes Stock introduces a public price history feature, allowing buyers and sellers to view past sale prices, bidding trends, and current market demand for any listed sneaker. This data-driven pricing model enables buyers to make informed purchasing decisions, ensuring that they are not overpaying due to price manipulation. Sellers, in turn, can set competitive asking prices that reflect real-time market value, promoting fair competition within the resale industry (Palmer, 2010).

5.3. Ethical user data handling

User privacy is a core ethical principle at Nepal Shoes Stock, ensuring that all personal information remains confidential and secure. In an era where data breaches and unauthorized data sharing have become major concerns, the platform upholds strict privacy standards, protecting users from potential exploitation. Many online marketplaces and e-commerce platforms profit from selling user data to third parties for targeted advertising or commercial gain, often without explicit consent. Nepal Shoes Stock rejects such unethical practices, guaranteeing that user data is never sold, shared, or misused for any external business interests. To reinforce ethical data handling, Nepal Shoes Stock implements robust encryption protocols and access controls, ensuring that user profiles, transaction histories, and payment details are stored securely. The platform follows international best practices for data security, despite Nepal's lack of comprehensive GDPR-style regulations. Additionally, users have the right to control their personal data, including options to modify, delete, or restrict the use of their information, fostering transparency and trust in the platform's privacy policies (Patrick E. Murphy, 2005).

5.4. Corporate social responsibility (CSR) initiatives

Nepal Shoes Stock is more than just a marketplace—it actively contributes to the growth and development of Nepal's sneaker community through various educational and

cultural initiatives. Recognizing that sneaker culture is still evolving in Nepal, the platform takes an active role in educating buyers, sellers, and enthusiasts about the history, authenticity, and value of sneakers. By organizing sneaker education programs, Nepal Shoes Stock helps users understand key aspects such as sneaker identification, authenticity verification, resale trends, and sneaker investment strategies. These programs are designed to empower both casual buyers and serious resellers, ensuring that the sneaker resale market in Nepal operates ethically and transparently. Additionally, collaborations with local artists and designers allow Nepal Shoes Stock to support and promote Nepali creativity in sneaker culture. By featuring custom sneaker designs, artist collaborations, and limited-edition releases, the platform provides exposure and economic opportunities for local talent. These initiatives help bridge the gap between international sneaker trends and Nepal's growing sneaker industry, fostering a unique and thriving local sneaker culture (Antonis Papasavva, 2024).

6. Security Aspects

6.1. Encryption and data protection

Nepal Shoes Stock prioritizes data security and user privacy by implementing AES 256 encryption, one of the most advanced encryption standards available. This ensures that all user data, payment details, and transaction information remain confidential and protected from potential cyber threats such as hacking, data breaches, and unauthorized access. Given the increasing risks associated with online transactions and digital identity theft, the platform adopts high-level security measures to create a safe and trustworthy environment for buyers and sellers. AES-256 encryption is widely recognized for its robust security capabilities, as it encrypts data in a way that makes it virtually impossible for hackers to decrypt without the proper authorization. Nepal Shoes Stock applies this encryption to all stored and transmitted data, ensuring that sensitive information—such as user credentials, payment transactions, and order details—remains protected from cybercriminals. Additionally, the platform employs Secure Socket Layer (SSL) technology, which provides an extra layer of security by encrypting all communications between users and the platform, preventing man-in-the-middle (MITM) attacks and data interception (Maryam Habibpour, 2023).

6.2. Secure payment processing

Nepal Shoes Stock ensures secure and reliable payment processing by integrating verified third-party payment gateways, minimizing the risks of fraud, unauthorized

transactions, and financial data breaches. In e-commerce and online marketplaces, payment security is a critical concern, as cybercriminals often exploit vulnerabilities in unsecured payment systems to commit fraud, steal credit card information, or manipulate transactions. To address these risks, Nepal Shoes Stock partners with trusted and well-established payment processors, ensuring that all transactions are encrypted, authenticated, and securely processed. By utilizing third-party payment gateways, the platform benefits from advanced fraud detection algorithms, which analyze transactions in real-time to identify suspicious activities such as unauthorized payments, account takeovers, and duplicate transactions. Additionally, multi-layered security protocols, including tokenization and 8 encryption, protect sensitive financial data by replacing real card details with encrypted tokens, preventing credit card fraud and data interception (gocardless, 2023).

6.3. Multi factor authentication (MFA) for account security

To prevent unauthorized access and enhance transaction security, Nepal Shoes Stock integrates multi-factor authentication (MFA), particularly for high-value transactions and sensitive user actions. Online marketplaces are often targeted by cybercriminals, identity thieves, and fraudsters, making it crucial to implement robust security measures that protect users from unauthorized account access and fraudulent purchases. MFA strengthens security by requiring users to verify their identity through multiple authentication steps rather than relying solely on a username and password, which can be easily compromised through phishing or hacking. Before completing a high-value sneaker transaction, users must authenticate their identity using additional security factors, such as one-time passwords (OTPs) sent via SMS or email, biometric authentication like fingerprint or facial recognition, or backup security questions. These added layers of verification ensure that even if a hacker obtains a user's password, they cannot access the account or complete unauthorized purchases without the secondary authentication step. Furthermore, Nepal Shoes Stock monitors login attempts and transaction patterns, flagging suspicious activities and prompting additional security verification if an unusual transaction is detected. By implementing MFA for critical actions, such as logins, withdrawals, and expensive purchases, Nepal Shoes Stock significantly reduces the risk of fraud, reinforcing its position as a secure and trustworthy sneaker marketplace where users can buy and sell with confidence and peace of mind (Anon., August 23, 2024).

6.4. fraud prevention and scam detection

To enhance security and maintain a trustworthy sneaker marketplace, Nepal Shoes Stock integrates AI-powered fraud detection to identify suspicious activities, block fraudulent users, and prevent scams before they occur. Fraudulent activities, such as fake listings, unauthorized transactions, and identity theft, are common risks in online resale markets, making it essential to adopt proactive security measures (Coderre, 2009). By leveraging artificial intelligence and machine learning algorithms, the platform continuously analyzes user behavior, transaction patterns, and seller activities to detect any anomalies that may indicate potential fraud. If a transaction or user activity appears suspicious, such as multiple failed login attempts, unusual bidding behavior, or a sudden change in selling patterns, the system automatically flags the account for further review. In cases where fraudulent behavior is confirmed, Nepal Shoes Stock immediately blocks the user, cancels the transaction, and prevents further access to the platform. This AI-driven approach allows for real-time fraud detection, significantly reducing the chances of buyers falling victim to scams or counterfeit sales. Additionally, the platform keeps a record of flagged accounts, preventing repeat offenders from re-registering under new identities. By implementing AI-powered fraud detection, Nepal Shoes Stock ensures a secure, scam-free environment, allowing sneaker enthusiasts to trade with confidence and peace of mind while protecting the integrity of the marketplace. (Anon., August 23, 2024)

7. Project Methodology

The development of **Nepal Stock Shoes (NSS)** followed the **Agile methodology**, which emphasizes iterative development and flexibility. The project was divided into seven distinct sprints, each focusing on specific tasks to ensure incremental progress. Agile allowed for continuous feedback and adjustments, ensuring the platform met user expectations and functional requirements.

Sprint 1: Planning and Designing

In the first sprint, the primary objective was to gather and analyze the project requirements. Extensive market research was conducted to understand the challenges faced by sneaker enthusiasts in Nepal. Based on this research, the key features of NSS were defined. Wireframes and mockups of the user interface were created to visualize the platform's structure. Additionally, the database schema and system architecture were designed using Entity-Relationship (ER) diagrams. This sprint also involved creating a detailed project plan, identifying milestones, and setting deadlines.

Sprint 2: User Management System

The second sprint focused on developing a secure **User Management System**. The primary tasks included implementing user registration, login, and logout functionalities using **PHP** and **MySQL**. User roles (buyer, seller, admin) were established with appropriate access controls. Additionally, password encryption and authentication mechanisms were integrated for secure access. Users were also provided with the option to reset their passwords.

Sprint 3: Product Management

In this sprint, the platform's **Product Management System** was built. Sellers were given the ability to list sneakers by providing details such as product name, brand, size, price, and images. Buyers could browse products using filtering and sorting options. The product editing and deletion functionalities were implemented to ensure sellers could manage their listings effectively. Admin users were also provided with controls to verify product authenticity and monitor listings.

Sprint 4: Sales Tracking

The fourth sprint involved implementing a comprehensive **Sales Tracking System**. Real-time tracking of sales data was introduced to ensure transparency.

Both sellers and buyers could view their sales history and track the status of ongoing transactions. A reporting feature was added to generate visual representations of market trends, helping sellers make informed pricing decisions. Additionally, sales analytics were made available to the admin for platform monitoring.

Sprint 5: Chat Support

A **Real-Time Chat System** was integrated in the fifth sprint using **JavaScript** and **WebSockets**. This feature allowed buyers and sellers to communicate directly, resolving queries and negotiating deals. A dedicated admin chat support system was also implemented to provide customer assistance and handle disputes. The chat system ensured secure communication and retained chat histories for future reference.

Sprint 6: Admin Control Panel

During this sprint, a robust **Admin Control Panel** was developed. Admin users were given access to a dashboard to monitor user activities, manage product listings, and resolve conflicts. The panel included functionalities for blocking suspicious users, removing counterfeit product listings, and reviewing reported issues. Fraud detection algorithms were integrated to flag unusual activities for further investigation.

Sprint 7: Testing and Evaluation

The final sprint was dedicated to thorough testing and evaluation of the platform. **Black-box testing** was performed to ensure each feature functioned as expected. Test cases were executed to validate user registration, product management, sales tracking, chat functionality, and admin operations. Any bugs

or errors identified during this phase were resolved promptly. User feedback was collected for further refinement, ensuring a smooth and error-free user experience.

Gantt Chart

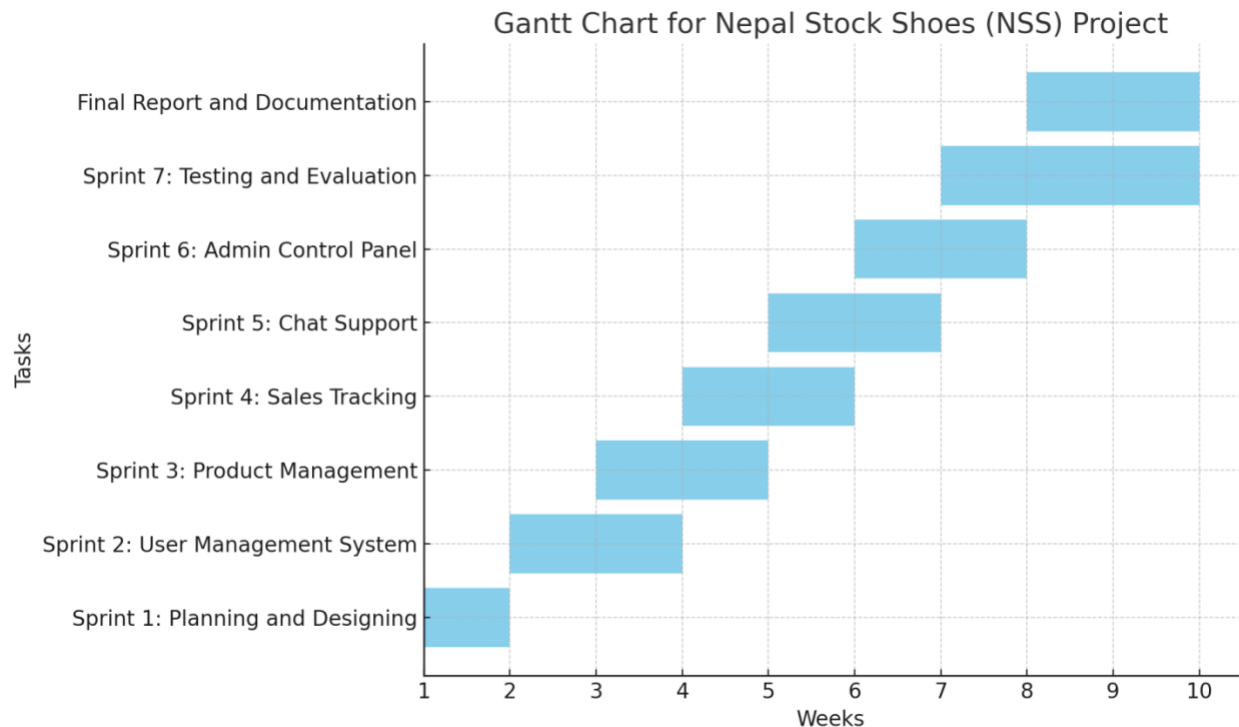


Figure 1 Gantt chart

8. Different Technology and Tools used for the project

The development of **Nepal Stock Shoes (NSS)** involved a combination of technologies and tools to ensure a seamless, secure, and efficient platform. Each technology was chosen for its specific strengths in supporting the project's requirements.

PHP

- **Purpose:** PHP was used as the primary server-side scripting language for developing the backend of the platform.

- **Role:** It handled the core logic of the platform, including user authentication, product management, chat functionality, and admin operations.

- **Features:**

- Facilitated secure communication between the frontend and the database.
- Managed business logic for transactions and user management.
- Integrated with MySQL for data storage and retrieval.
- Supported form handling for user input validation.

- **Why PHP?** It is a versatile, widely supported language ideal for building dynamic web applications, especially those requiring database connectivity.

Visual Studio Code (VS Code)

- **Purpose:** Visual Studio Code was used as the primary Integrated Development Environment (IDE) for writing and managing the project's codebase.

- **Role:** It provided a user-friendly environment for code editing, debugging, and testing.

- **Features:**

- Integrated support for PHP, JavaScript, and MySQL with helpful extensions.
- Built-in debugging tools to troubleshoot code efficiently.
- Git integration for version control and collaboration.
- Live server extension to test web applications locally.

- **Why VS Code?** Its lightweight nature, extensive library of extensions, and built-in development tools made it an ideal choice for managing both frontend and backend development.

MySQL

- **Purpose:** MySQL served as the relational database management system for storing and managing the platform's data.

- **Role:** It stored user data, product information, sales transactions, chat messages, and admin records.

- **Features:**

- Provided secure and reliable data storage.

- Allowed complex queries to retrieve data efficiently.
- Supported data integrity using relationships, constraints, and normalization.
- Enabled easy data management through SQL queries.
- **Why MySQL?** It is a robust, scalable, and open-source database management system that seamlessly integrates with PHP, making it a perfect choice for web applications requiring structured data management.

These technologies and tools collectively contributed to the development of a functional, secure, and user-friendly sneaker resale platform, ensuring a seamless experience for all users on **Nepal Stock Shoes (NSS)**.

9. Artefact Designs:

User Management System (UMS)

Req.Code	Requirement Description	MoSCoW prioritization
UMS-F-1.0	The system should allow buyers, sellers and admins to register and log in.	Must have
UMS-F- 1.1	The User passwords should be securely encrypted.	Must Have
UMS-F- 1.2	Admins should be able to create, disable, and remove user accounts.	Must Have
UMS-F 1.3	The system should have an intuitive user profile management interface.	Must Have
UMS-F 1.4	The system should display error messages when login issues occur.	Must Have
UMS NF-1.5	The system should back up user data regularly for recovery.	Should Have

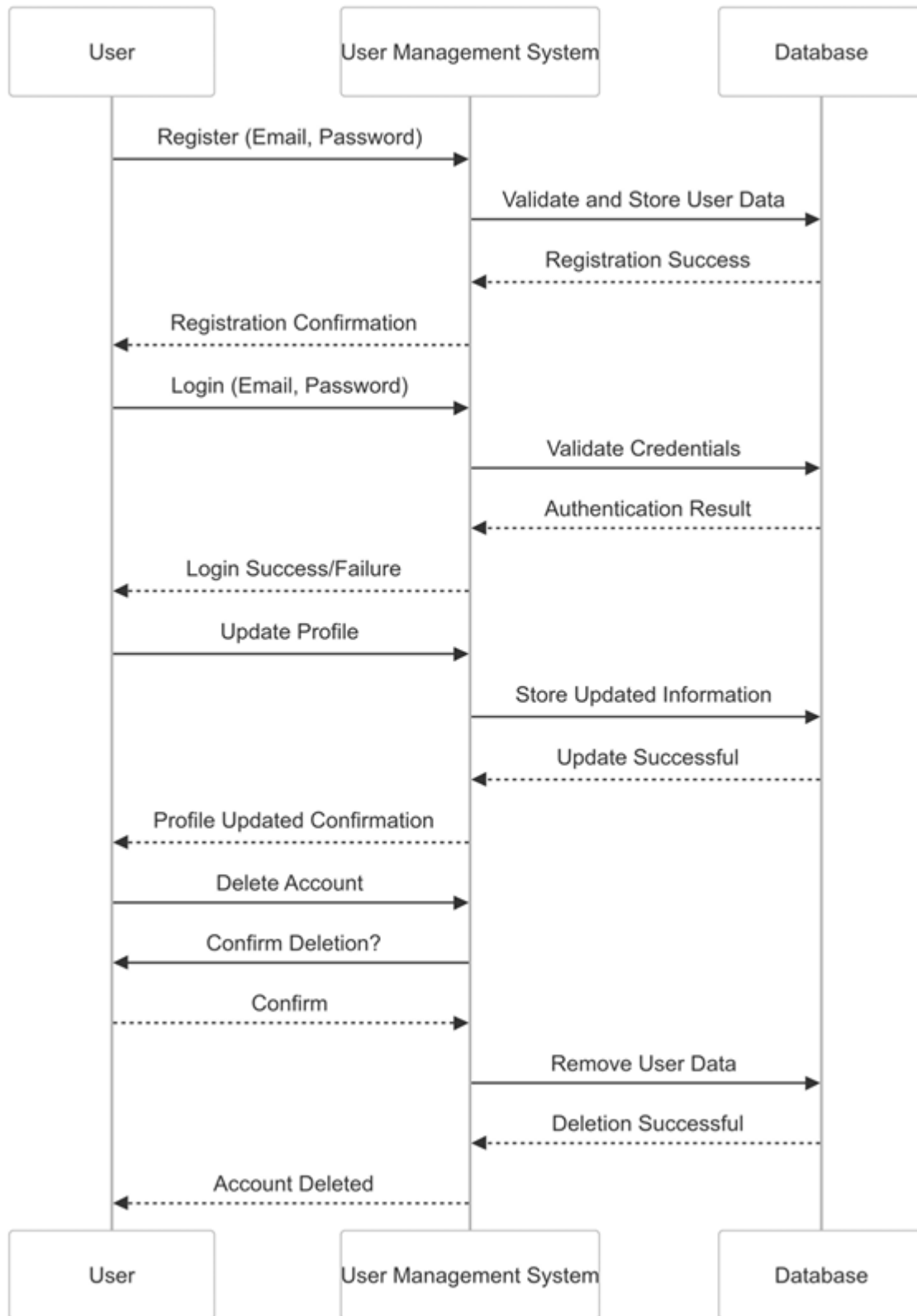


Figure 2 User management system sequence diagram

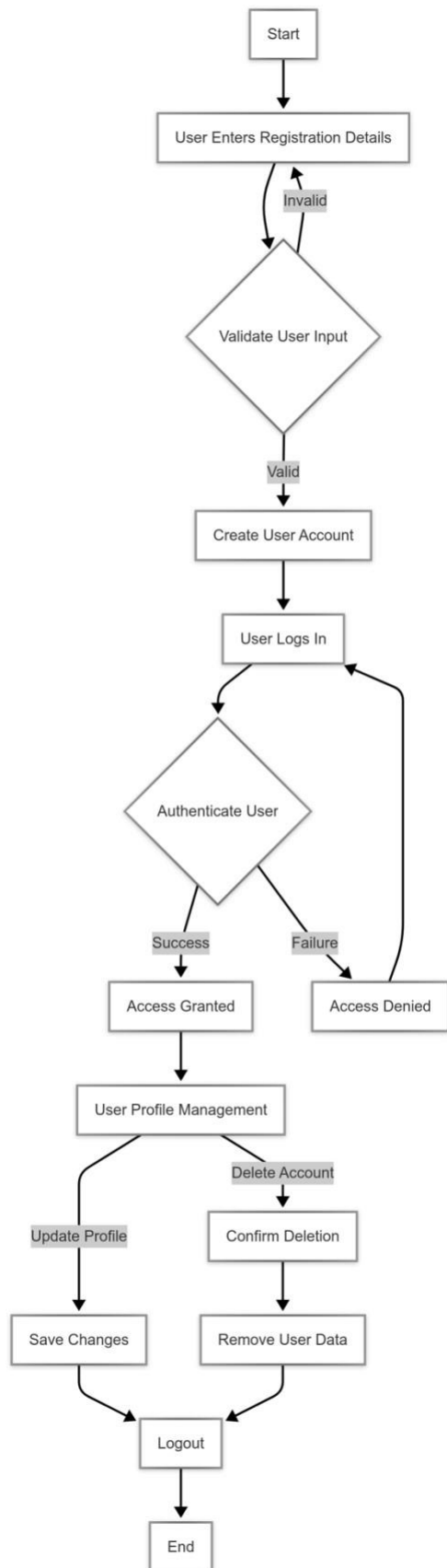


Figure 3 Activity diagram

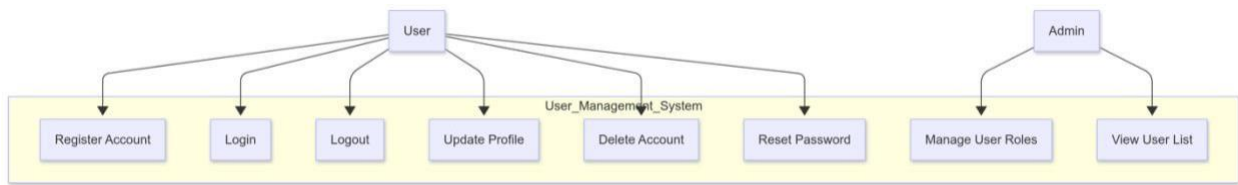


Figure 4 User management system

Product Management System(PTS)

Req. Code	Requirement Description	Prioritization
PMS-F- 1.0	Sellers should be able to add, edit, and delete products.	Must Have
PMS-F- 1.1	Products should have categories, tags, and high-quality images.	Must Have
PMS-F- 1.2	The system should track stock availability.	Must Have
PMS-F- 1.3	The product listing page should have a search and filtering system.	Must Have
PMS-F- 1.4	The system should generate graphical reports of symptoms.	Must Have

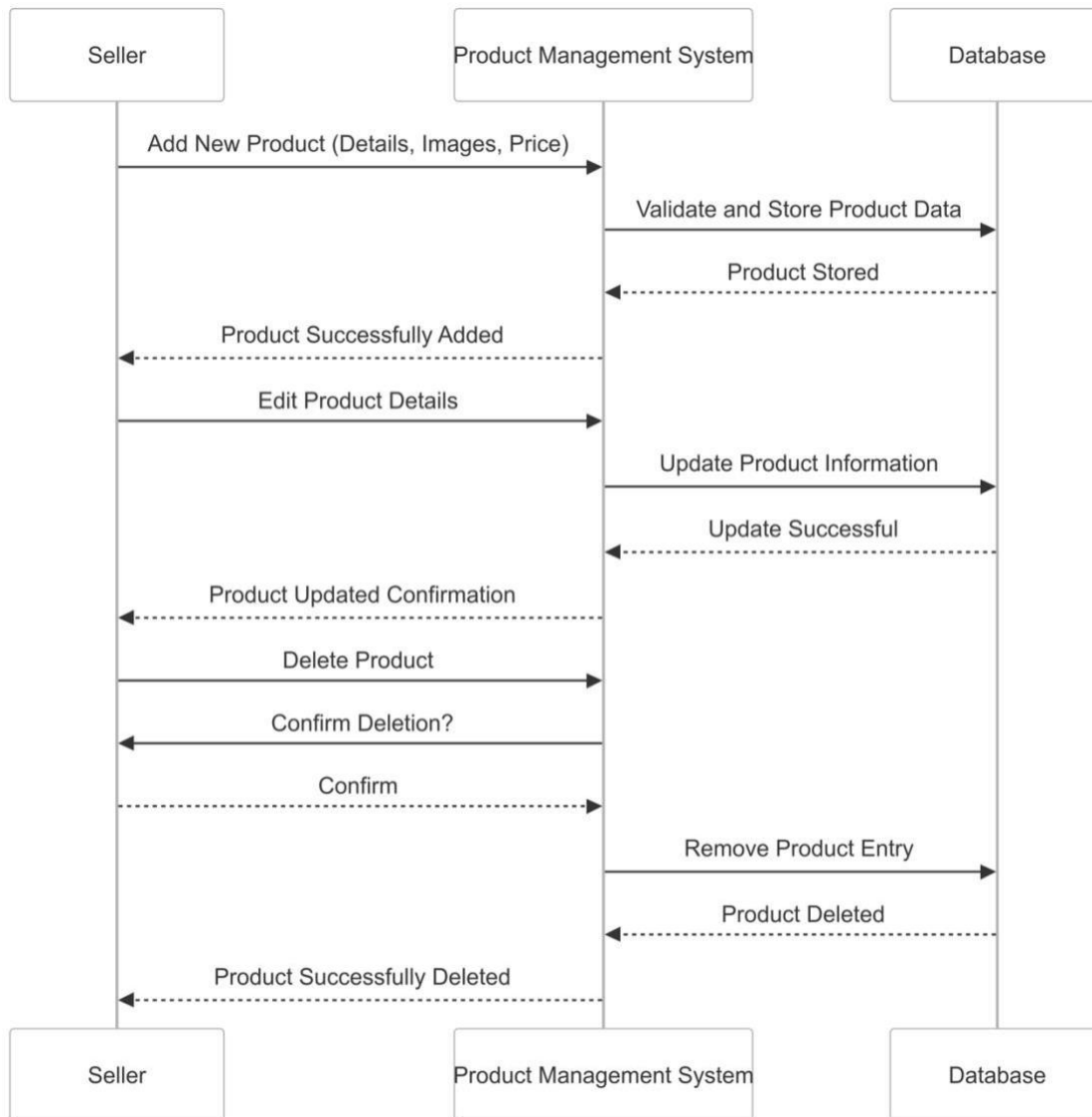


Figure 5 Sequence diagram

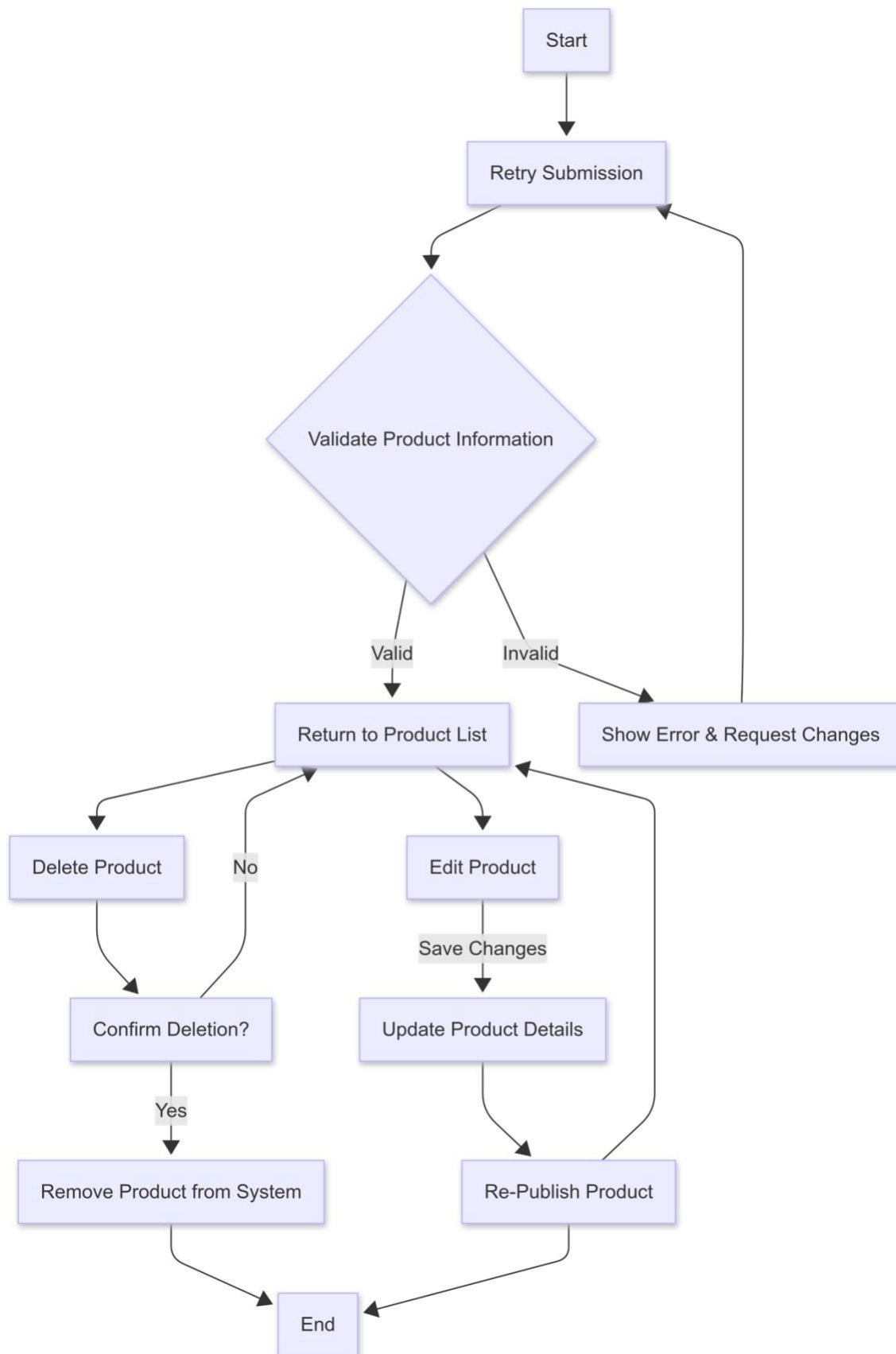


Figure 6 Activity diagram

Dynamic pricing System(DPS)

Req. Code	Requirement Description	Prioritization
DPS-F- 1.0	The system should dynamically adjust prices based on supply and demand.	Must Have
DPS-F- 1.1	The system should fetch real-time product demand analytics.	Must Have
DPS-F- 1.2	Price changes should be logged for transparency.	Must Have
DPS-F- 1.3	Users should be notified about price fluctuations.	Should Have
DPS-F- 1.4	The system should update prices within 5 seconds of demand shifts.	Should Have

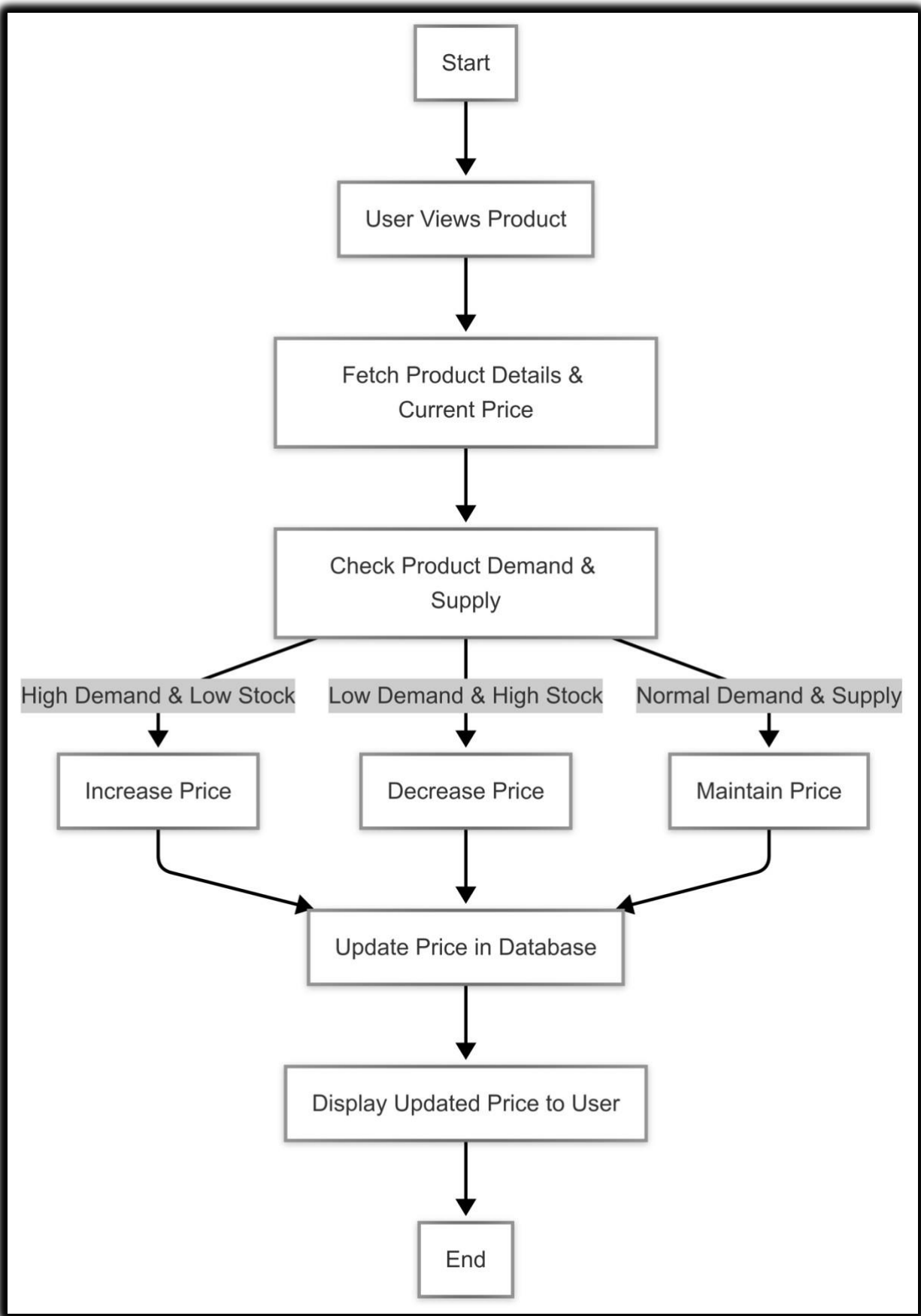


Figure 7 Activity diagram

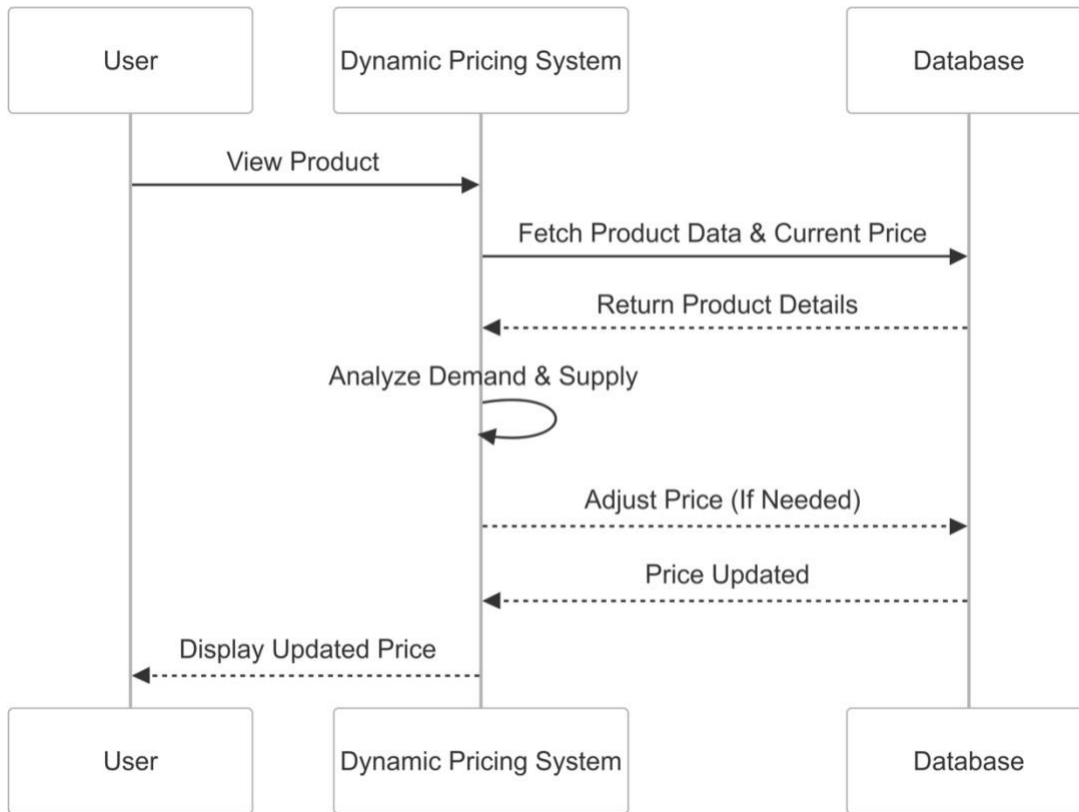


Figure 8 Sequence diagram

Order & Checkout Subsystem (OCS)

Req. Code	Requirement Description	Prioritization
OCS-F- 1.0	Buyers should be able to add products to a shopping cart.	Must Have
OCS-F- 1.1	Users should select a shipping address during checkout.	Must Have
OCS-F- 1.2	Users should review and confirm their order before payment.	Must Have
OCS-F- 1.3	The checkout page should be simple and user-friendly.	Must Have
OCS-F- 1.4	Orders should be processed within 2 seconds.	Should Have

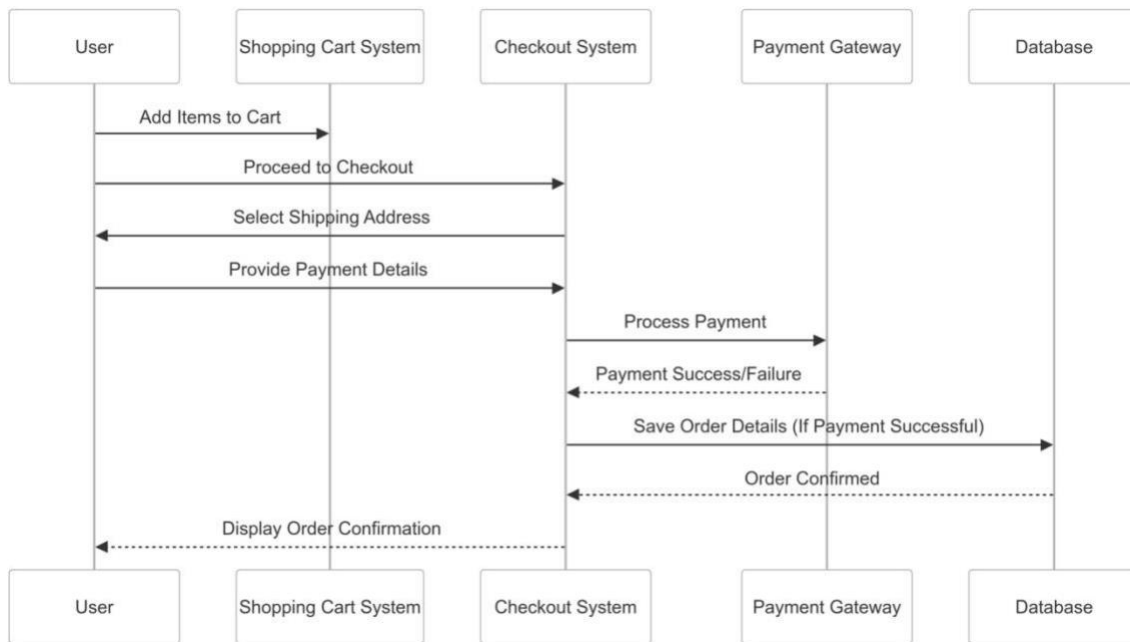


Figure 9 Sequence diagram

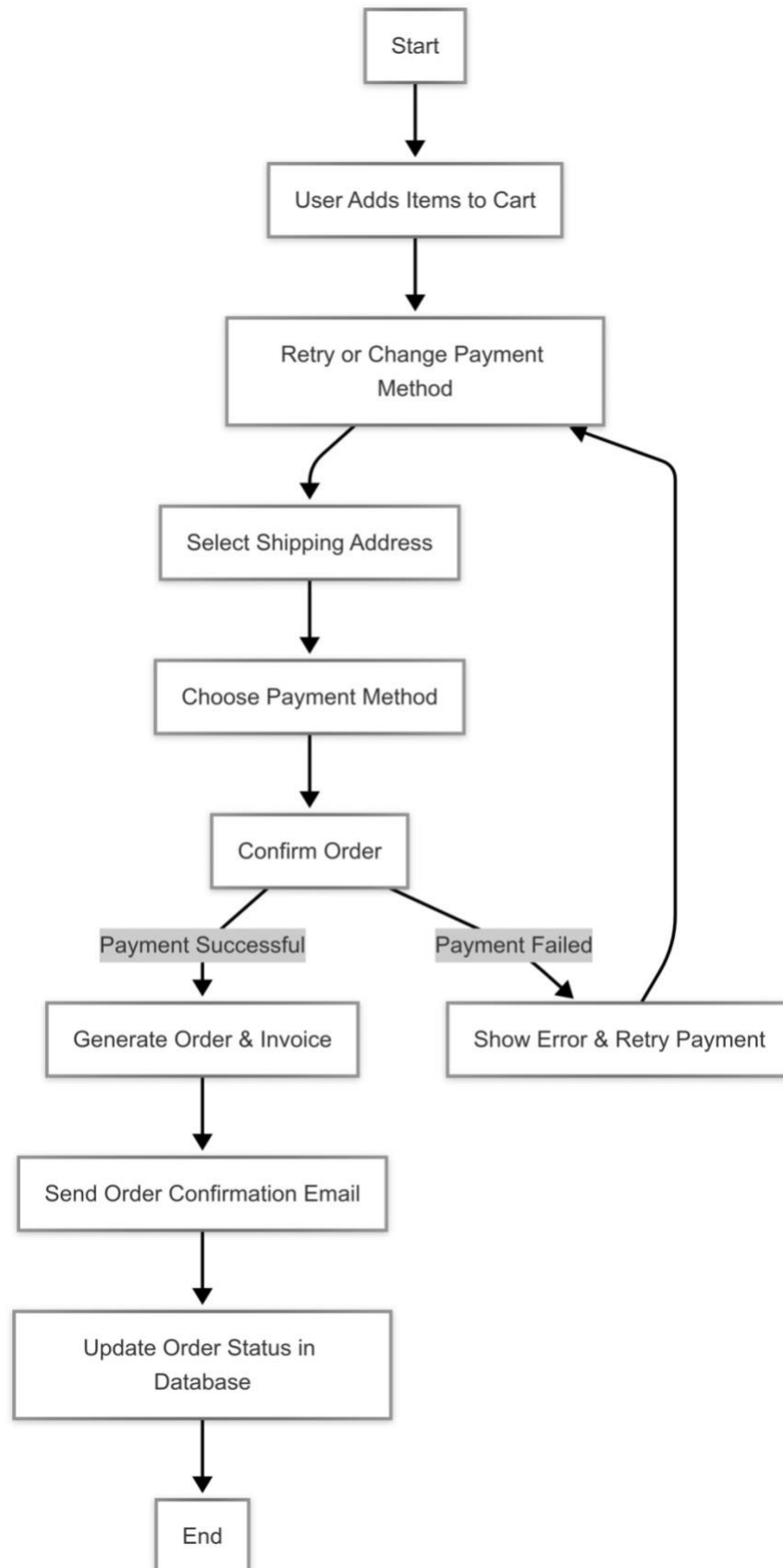
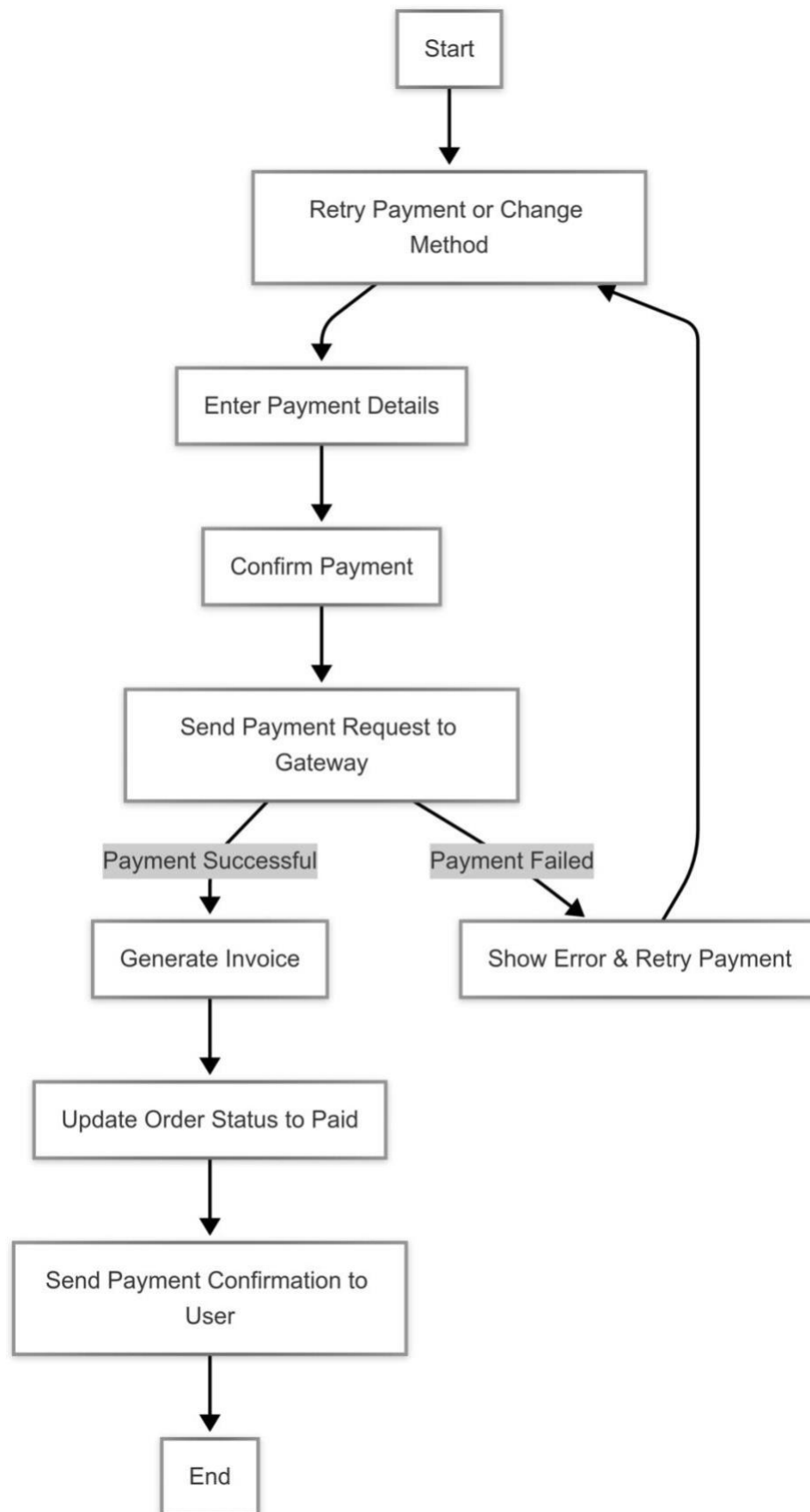


Figure 10 Activity diagram

Payment & Transaction Subsystem

Req. Code	Requirement Description	Prioritization
PTF-F- 1.0	Buyers should be able to make secure payments.	Must Have
PTF-F- 1.1	The system should integrate multiple payment gateways (PayPal, Stripe, UPI).	Must Have
PTF-F- 1.2	Payment failures should trigger an error message.	Must Have
PTF-F- 1.3	Users should be able to request refunds if applicable.	Must Have
PTF-F- 1.4	Payment transactions should be completed in under 3 seconds.	Should Have



Wire Frames

1. Login Page

- Simple and clean design.
- Fields for **Username/Email** and **Password**.
- **Login** and **Forgot Password** options.
- A button to navigate to the **Sign-Up** page.
- Error messages displayed in case of incorrect login attempts.

2. Registration Page

- Fields for **Full Name**, **Email**, **Password**, and **Confirm Password**.
- User role selection (**Buyer** or **Seller**).
- **Register** button to create an account.
- Option to navigate back to the **Login** page.

3. Home Page

- Displays featured and popular sneakers.
- Search bar for finding specific sneakers.
- Filter and sort options for easy browsing.
- **Sign In** and **Sign Up** buttons for non-logged-in users.
- Logged-in users can access their dashboard.

4. Product Listing Page

- Displays sneaker listings in a grid view.
- Product cards with images, names, sizes, and prices.
- **View Product** button for more details.
- Search and filter functionality for refining results.

5. Product Details Page

- Detailed view of the sneaker with multiple images.
- Description of the product, including size, brand, and condition.
- Price, bidding options, and sales history.
- **Buy Now** or **Place Bid** buttons.
- Chat option for buyer-seller communication.

6. User Dashboard (Buyer/Seller)

- **Sellers** can manage their listings, add new products, and track sales.
- **Buyers** can view their purchase history and track active bids.
- Notification center for important updates.
- Access to the chat feature for communication.

7. Admin Dashboard

- Overview of platform activity with visual analytics.
- Manage users and product listings.
- Monitor transactions and resolve disputes.
- Fraud detection alerts for suspicious activity.

8. Chat Page

- Real-time chat between buyers and sellers.
- Option to contact platform support for assistance.
- Simple and clean messaging interface.



The wireframe illustrates a login page layout. At the top, a light gray rectangular header contains the text "Login Page". Below this header, there are two vertically stacked white rectangular input fields, each labeled "Input Field" in its top-left corner. Centered below the input fields is a solid blue rectangular button labeled "Button". At the bottom of the form, centered, is a blue hyperlink labeled "Forgot Password?".

Figure 12 Wireframe 1

Product Listing Page

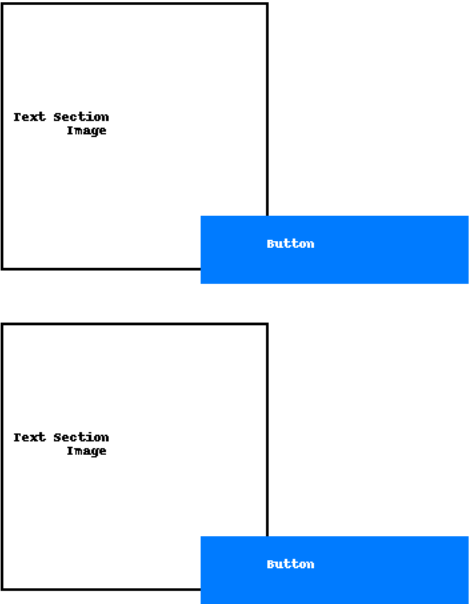


Figure 13 Wireframe 2

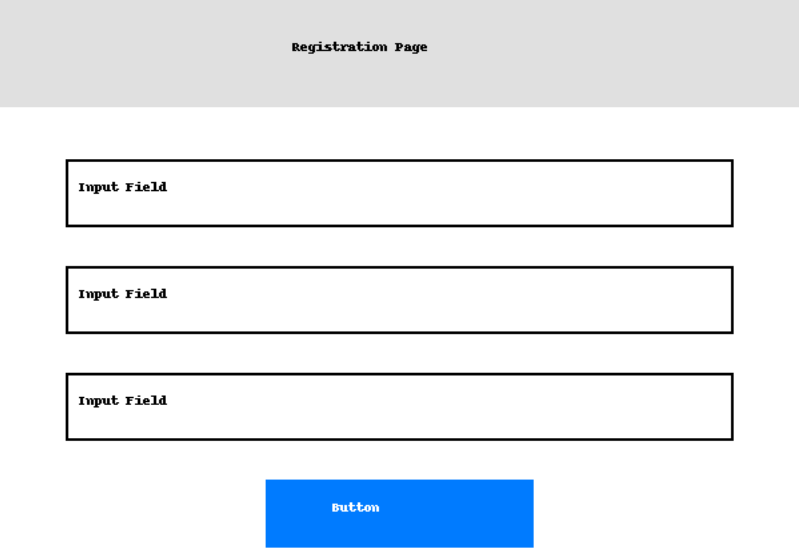


Figure 14 Wireframe 3

Class Diagram

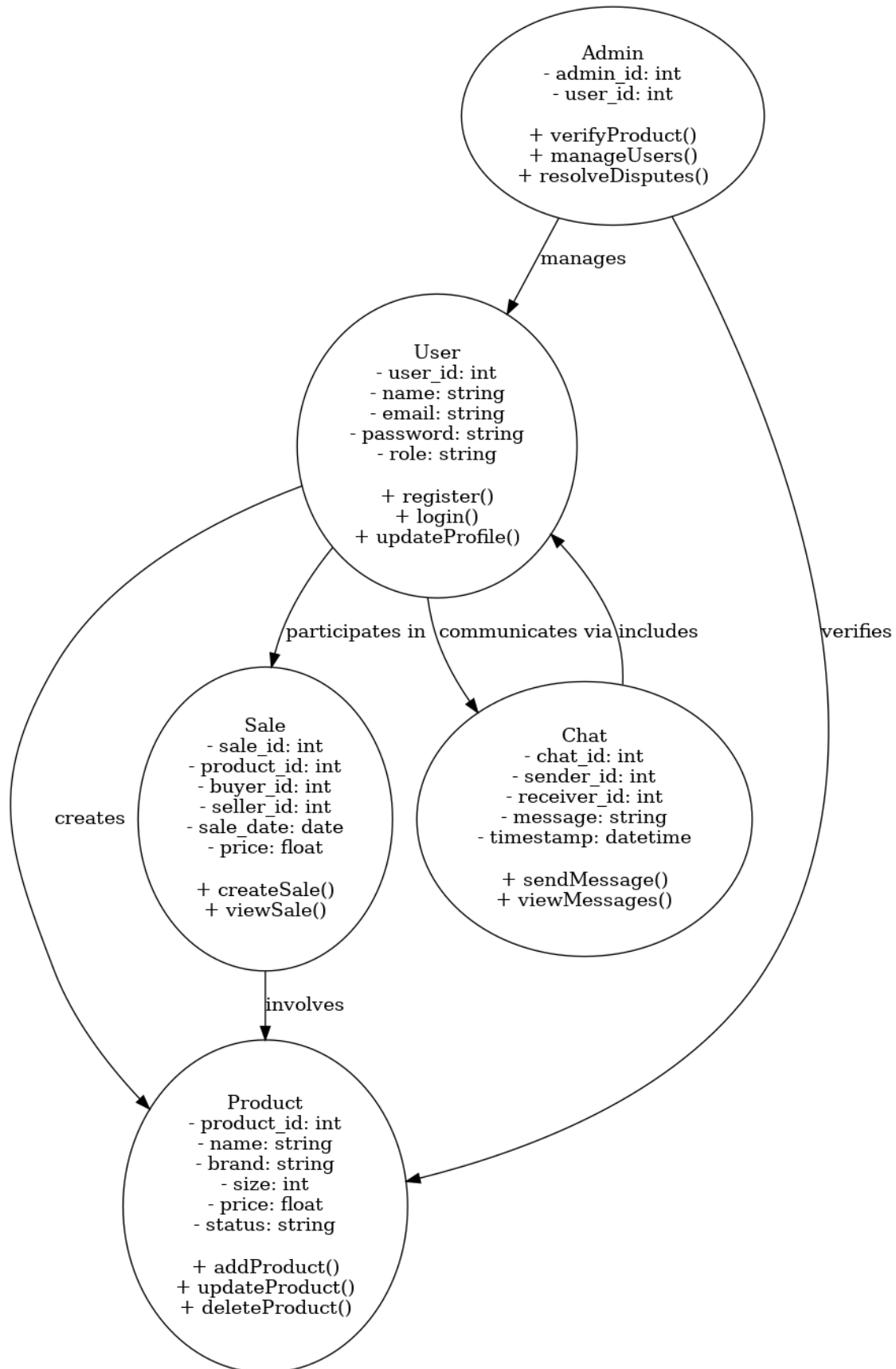


Figure 15 Class diagram

ER Daigram

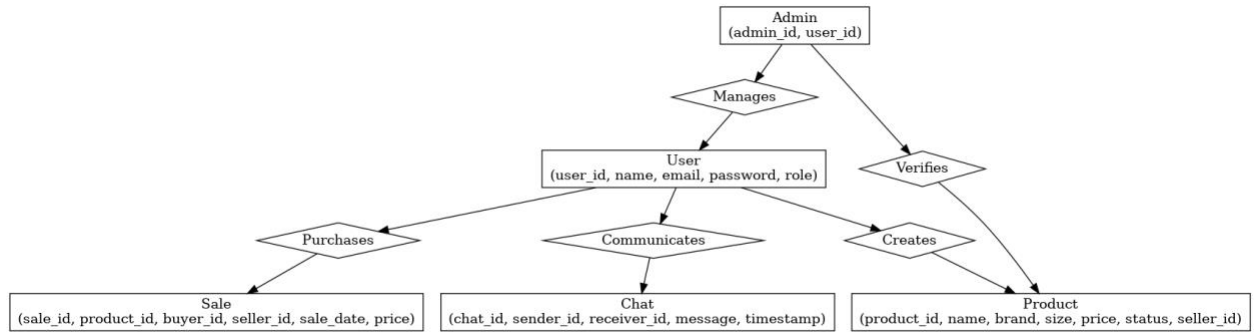


Figure 16 ER diagram

System Testing

Test ID	Test Description	Input	Expected Output	Actual Output	Status
TC-01	User Registration with valid details	Name, Email, Password, Role	Account created successfully	Account created successfully	☑ Passed
TC-02	User Registration with existing email	Existing Email, Password	Error: Email already registered	Error: Email already registered	☑ Passed
TC-03	Login with correct credentials	Valid Email, Password	Login successful	Login successful	☑ Passed
TC-04	Login with incorrect credentials	Invalid Email, Password	Error: Invalid credentials	Error: Invalid credentials	☑ Passed
TC-05	Product Listing by Seller	Product Name, Brand, Size, Price	Product added successfully	Product added successfully	☑ Passed
TC-06	View Product Details	Product ID	Product details displayed	Product details displayed	☑ Passed
TC-07	Purchase Product	Buyer ID, Product ID, Payment	Purchase completed, Sale recorded	Purchase completed	☑ Passed
TC-08	Chat Initiation Between Buyer and Seller	Buyer ID, Seller ID, Message	Message sent successfully	Message sent successfully	☑ Passed
TC-09	Admin Verification of Product	Admin ID, Product ID	Product marked as verified	Product marked as verified	☑ Passed
TC-10	Password Reset	Registered Email	Password reset link sent	Password reset link sent	☑ Passed

Test Case

Here is a detailed test case format for the **Nepal Stock Shoes (NSS)** platform. Each test case includes the test case ID, description, preconditions, test steps, expected results, actual results, and status.

Test Case 1: User Registration with Valid Dtails

- **Test Case ID:** TC-01
- **Description:** Validate that users can successfully register with valid details.

- **Preconditions:** User is on the registration page.

- **Test Steps:**

1. Enter valid **Name**, **Email**, **Password**, and **Confirm Password**.

2. Select user role (Buyer/Seller).

3. Click the **Register** button.

- **Expected Result:** The user account is created successfully, and a confirmation message is displayed.

- **Actual Result:** Account created successfully.

- **Status:**  Passed

Test Case 2: User Registration with Existing Email

- **Test Case ID:** TC-02

- **Description:** Verify that registration fails when using an already registered email.

- **Preconditions:** The email is already registered in the system.

- **Test Steps:**

1. Enter the registered **Email**.

2. Enter valid **Name** and **Password**.

3. Click the **Register** button.

- **Expected Result:** The system should display an error message indicating that the email is already registered.

- **Actual Result:** Error message displayed.

- **Status:**  Passed

Test Case 3: Login with Correct Credentials

- **Test Case ID:** TC-03
- **Description:** Verify that users can log in using correct credentials.
- **Preconditions:** User must be registered.
- **Test Steps:**
 1. Enter valid **Email** and **Password**.
 2. Click the **Login** button.
- **Expected Result:** User successfully logs in and is redirected to the dashboard.
- **Actual Result:** Login successful.
- **Status:**  Passed

Test Case 4: Login with Incorrect Credentials

- **Test Case ID:** TC-04
- **Description:** Ensure the system blocks login attempts with incorrect credentials.
- **Preconditions:** User enters incorrect login details.
- **Test Steps:**
 1. Enter invalid **Email** or **Password**.
 2. Click the **Login** button.

- **Expected Result:** An error message should appear stating that the credentials are incorrect.
- **Actual Result:** Error message displayed.
- **Status:**  Passed

Test Case 5: Product Listing by Seller

- **Test Case ID:** TC-05
- **Description:** Validate that sellers can list a product on the platform.
- **Preconditions:** Seller is logged in.
- **Test Steps:**
 1. Go to the seller dashboard.
 2. Enter valid product details (Name, Brand, Size, Price, Image).
 3. Click the **Add Product** button.
- **Expected Result:** Product is listed successfully and visible in the product catalog.
- **Actual Result:** Product added successfully.
- **Status:**  Passed

Test Case 6: Chat Functionality

- **Test Case ID:** TC-06
- **Description:** Verify that buyers and sellers can communicate using the chat system.
- **Preconditions:** Buyer and seller accounts exist, and the product is listed.

- **Test Steps:**

1. Buyer selects a product and clicks on **Chat with Seller**.
2. Sends a message using the chat interface.
3. Seller responds to the message.

- **Expected Result:** Messages should be delivered and displayed in real-time.

- **Actual Result:** Messages exchanged successfully.

- **Status:**  Passed

10. Conclusion

The **Nepal Stock Shoes (NSS)** project successfully addresses the challenges faced by sneaker enthusiasts in Nepal's resale market by providing a transparent, secure, and user-friendly platform. Through its key features like secure user management, real-time sales tracking, product authentication, and live chat support, NSS enhances the sneaker trading experience for both buyers and sellers.

The project adopted an **Agile development methodology**, allowing flexibility and continuous improvement throughout the implementation phases. Systematic testing ensured that the platform met the functional, security, and performance requirements. All major components, including product management, sales

tracking, user authentication, and administrative controls, performed as expected, ensuring a smooth user experience.

Security and Transparency were the core priorities of the NSS platform. Secure payment integration, fraud detection mechanisms, and product verification processes minimize the risk of counterfeit products and fraudulent activities. Additionally, the subscription-based business model provides sellers with a cost-effective way to reach a wider audience.

Furthermore, the incorporation of an **Admin Control Panel** allows platform administrators to manage users, monitor transactions, and address disputes efficiently. Real-time communication through the chat support system also enhances user engagement and satisfaction.

While the current version of NSS is designed to serve the Nepalese market, future expansions can include:

- **International Expansion:** Expanding the platform's reach to other regions.
- **Mobile Application:** Developing a mobile version for convenient access.
- **AI-Powered Authentication:** Implementing AI for advanced counterfeit detection.
- **Additional Payment Options:** Integrating international payment gateways.

In conclusion, **Nepal Stock Shoes (NSS)** has successfully established itself as a reliable and innovative platform for sneaker resale in Nepal. The project demonstrates the effectiveness of using technology to solve real-world problems, fostering trust and transparency in the sneaker market. Future enhancements will further strengthen its position as the go-to platform for sneaker enthusiasts in Nepal and beyond.

11. References

Competitive Pricing on Online Markets: A Literature Review [Book] / auth. Berends Torsten J. Gerpott and Jan. - [s.l.] : he Journal of Revenue and Pricing Management , 2022.

Ethics for the Information Age (8th Edition) [Book] / auth. Quinn Michael J.. - New York : Pearson, 2020.

Ethical Issues in E-Business: Models and Frameworks [Book] / auth. Palmer Daniel E.. - Hershey, PA : Business Science Reference, 2010.

Ethical Marketing [Book] / auth. Patrick E. Murphy Gene R. Laczniak, Norman E. Bowie, Thomas A. Klein. - Upper Saddle River, NJ : Pearson, 2005.

Fraud Detection: A Revealing Look at Fraud [Book] / auth. Coderre David G.. - Vancouver, Canada : Global Audit Publications, 2009.

Application of AI-based Models for Online Fraud Detection and Analysis [Book] / auth. Antonis Papasavva Shane Johnson, Ed Lowther, Samantha Lundrigan, Enrico Mariconti, Anna Markovska, Nilufer Tuptuk. - online : arXiv, 2024.

Uncertainty-Aware Credit Card Fraud Detection Using Deep Learning [Book] / auth. Maryam Habibpour Hassan Gharoun, Mohammadreza Mehdipour, AmirReza Tajally, Hamzeh Asgharnezhad, Afshar Shamsi, Abbas Khosravi, Miadreza Shafie-Khah, Saeid Nahavandi, Joao P. S. Catalao. - online : arXiv, 2023.

Designing Online Marketplaces: Trust and Reputation Mechanisms [Book] / auth. Luca Michael. - Chicago : University of Chicago Press, 2017.

Designing Online Marketplaces: Trust and Reputation Mechanisms [Book] / auth. Luca Michael. - 2017 : University of Chicago Press, Chicago.

Patent, Design, and Trademark Act [Book]. - nepal : Ministry of Industry, Commerce, and Supplies, Government of Nepal, 2022.